RICCO CABRAL VENTEREA

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EDUCATION

PhD University of California, Riverside *Aug 2025 – Present*
Astronomy

Università degli Studi di Perugia *Oct 2024 – Jun 2025*
Visiting Fulbright Research Scholar

BA Cornell University *Aug 2020 – May 2024*
Astronomy with Astrophysics Concentration *with honors: magna cum laude*
Minor in Physics

RESEARCH EXPERIENCE

Department of Physics and Astronomy, University of California, Riverside
Aug 2025 – Present
Center for Experimental Cosmology and Instrumentation
Graduate Student Researcher
Advisors: Professor Steve Choi, Dr. Junna Sugiyama
- Calculate angular power spectra of the cosmic microwave background and constrain secondary anisotropies from cosmology observatories.

Department of Physics and Geology, University of Perugia, Perugia, Italy
Oct 2024 – Jun 2025
Fulbright Research Scholar
Advisors: David Pelosi, Professor Nicola Tomassetti
- Created database for solar properties measured by global observatories.
- Developed cross-correlation study between solar modulation and solar proxies.

Cornell Center for Astrophysics and Planetary Science, Ithaca, NY
Jan 2024 – May 2024
Student Research Assistant III
Advisors: Professor Nicholas Battaglia, Dr. John Orlowski-Scherer
- Developed database containing asteroid thermal emission flux measurements using AWS.

School of Physics and Astronomy, University of Minnesota, Twin Cities, Minneapolis, MN
Jun 2023 – Sep 2024
LIGO Scientific Collaboration

A3D3 AI Institute
Research Assistant
Advisors: William Benoit, Professor Michael Coughlin

- Work in implementing machine learning algorithms to clean and analyze gravitational wave strain data collected from the Laser Interferometer Gravitational-Wave Observatory Livingston and Hanford sites.
- Generated constant-Q transforms for high signal-to-noise gravitational wave event simulations.
- Developed a user-interface to create distributions of black hole parameter simulations.

Nexus Scholars Program, Cornell University, Ithaca, NY *May 2022 – Jul 2022*
Research Assistant
Advisor: Professor Nicholas Battaglia

- Continued full-time research in measuring thermal emission of objects in the asteroid belt.
- Detected large asteroids with a high degree of confidence and generated light curves. Presented results at a symposium.

Department of Astronomy, Cornell University, Ithaca, NY *Sep 2021 – May 2024*
Research Assistant
Advisors: Professor Nicholas Battaglia, Dr. John Orlowski-Scherer

- Determined flux measurements from thermal emission data of objects in the asteroid belt. Data collected by the Atacama Cosmology Telescope.

National Aeronautics and Space Administration, Remote *May 2021 – Aug 2021*
Lucy Student Pipeline Accelerator and Competency Enabler
Mission Concept Academy
Astrophysicist

- Submitted a Preliminary Design Review on a science reconnaissance mission for water ice mapping in the lunar South Polar Region.

Summer Student Theoretical Physics Research Session, Remote *Jun 2021*
Student Researcher
Advisors: Professors Jim Gates, Kory Stiffler, Konstantinos Koutrolikos

- Learned topics in supersymmetry with a mathematical emphasis.
- Covered areas in group theory, Lie algebra, differential geometry, tensor algebra, Lagrangian dynamics, Clifford algebra, gauge theory, and supersymmetry algebra.
- Gained experience in LaTeX.

National Aeronautics and Space Administration, Remote *Jan 2021 – Apr 2021*
Lucy Student Pipeline Accelerator and Competency Enabler
Proposal Writing and Evaluation Experience Academy
Technical Team Member

- Submitted a research proposal on human health and performance in low-earth orbit.

Irondale High School, New Brighton, MN *Sep 2019 – May 2020*

Student Researcher
Advisor: Shane Wood
- Verified Einstein's Special Theory of Relativity using previous QuarkNet cosmic ray muon flux studies.
- Studied the effect of zenith angle on cosmic ray muon flux using the QuarkNet Cosmic Ray Muon Detector. Data collected at Irondale High School.
- Proposed and conducted an experiment in cosmic ray muons.

TEACHING EXPERIENCE

University of California, Riverside, Riverside, CA *Mar 2026 – Jun 2026*
Graduate Teaching Assistant, Department of Physics and Astronomy
- Teaching assistant for PHYS 2LA: General Physics Laboratory, a course covering the basic principles of classical mechanics for undergraduates.
- Wrote and graded ~ 35 weekly quizzes and laboratory reports.
- Supervised undergraduates in laboratory settings and proctored final laboratory skills assessment.

University of California, Riverside, Riverside, CA *Sep 2025 – Dec 2025*
Graduate Teaching Assistant, Department of Physics and Astronomy
- Teaching assistant for PHYS 40A: General Physics, a course introducing classical mechanics to undergraduates.
- Wrote and graded ~ 50 weekly quizzes and laboratory reports.
- Supervised undergraduates in laboratory settings and proctored final laboratory skills assessment.

PUBLICATIONS

Journal Publications

Marx E., Benoit W., Gunny A., et al., Machine-learning Pipeline for Real-time Detection of Gravitational Waves from Compact Binary Coalescences. 2025, PRD, 111

Orlowski-Scherer J., **Venterea R.**, Battaglia N., et al., The Atacama Cosmology Telescope: Millimeter Observations of a Population of Asteroids or: ACTeroids. 2024, ApJ, 964, 2

Venterea R., Orlowski-Scherer J., Næss S., et al., Sub-Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2023, American Astronomical Society Meeting Abstracts, 55, 2:104.12

Venterea R., Ekka U., An Introduction to Quantum Computing. 2022, JURP, 31, 1

Journal Papers in Review

Venterea R.C., Orlowski-Scherer J., Battaglia N., et al., The Atacama Cosmology Telescope: Release of A databaSe of millimeTer ObservatioNs of Asteroids Using acT (ASTRONAUT). 2025, ApJS

Campolongo E., Chou Y.T., Govorkova E., et al., Building Machine Learning Challenges for Anomaly Detection in Science. 2025, Nat. Commun.

Venterea R., Ekka U., An Analysis of Muon Flux from Angle Variation of the QuarkNet Cosmic Ray Detector. 2023, TPT

Conference Papers

(Peer-Reviewed)

Pelosi, D., Barão, F., Bertucci, B., et al., Modeling the Heliospheric Modulation of Cosmic Rays in Light of New Data from AMS-02 in Space. 2025, PoS

Pelosi, D., Barão, F., Bertucci, B., et al., Analysis of Galactic Cosmic Ray Variability and Time-Lagged Relation to Solar Activity. 2025, PoS

Reports

Venterea R. C., Predictions of the Solar Modulation Potential of Cosmic Rays Using Sunspot Number. 2025, U.S. Student Fulbright Program

Doku F., Ekka U., **Venterea R**., Executive Summary for the Spread of Misinformation and Levels of Censorship. 2021

Muralidhar A., Medvec M., Fujishima B., et al., Preliminary Design Review - Ad Lunam Hopper. 2021, National Aeronautics and Space Administration Lucy Student Pipeline Accelerator and Competency Enabler Mission Concept Academy

Mota A., Sin J., Garcia S., et al., Astronaut-Friendly 3 in 1 Edible Cutlery to Promote Bone Health. 2021, National Aeronautics and Space Administration Lucy Student Pipeline Accelerator and Competency Enabler Proposal Writing and Evaluation Experience Academy

Manuscripts

He Y., **Venterea R.**, Wu X., Food Distribution by Mobile Food Pantries: A Design for Optimized Schedule. 2020

PRESENTATIONS AND INVITED LECTURES

Presentations

Barão F., Bertucci B., Fiandrini E., et al., Analysis of Time Variability of Galactic Cosmic Rays. 2025, International Cosmic Ray Conference 2025

Venterea R., Orlowski-Scherer J., Battaglia, N., Næss S., et al., Sub-Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2024, Emerging Researchers in Exoplanet Science Symposium IX

Venterea R., Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2024, Cornell Undergraduate Research Board Spring Symposium

Venterea R., Galaxies and Asteroids. 2024, 13th Semi-Annual CURBx Research Conference

Marx E., Benoit W., Gunny A., et al., A search for binary mergers in archival LIGO data using aframe, a machine learning detection pipeline. 2024, April Meeting

Benoit W., Marx E., Gunny A., et al., A machine-learning pipeline for real-time detection of gravitational waves from compact binary coalescences. 2024, April Meeting

Venterea R., Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2023, Cornell University Undergraduate Research Poster Forum

Venterea R., Millimeter Observations of Asteroids Using the Atacama Cosmology Telescope. 2023, Cornell Undergraduate Research Board Spring Symposium

Venterea R., Observations of Asteroids with ACT. 2023, University of Rochester 2nd Annual Undergraduate Astronomy Research Seminar

Venterea R., Asteroids and ACT. 2022, Atacama Cosmology Telescope Collaboration Meeting

Venterea R., Looking at Asteroids. 2022, Cornell University Nexus Scholars Program Capstone Presentations

Venterea R., Curvature. 2021, Cornell University DRP Talks

Doku F., Ekka U., **Venterea R**., Problem C: Submitted a Tweet, Now What? 2021, SIMIODE Challenge Using Differential Equations Modeling

Venterea R., A Brief Introduction to Quantum Field Theory. 2021, Cornell University DRP Talks

Muralidhar A., Medvec M., Fujishima B., et al., Team 21 - Ad Lunam. 2021, National Aeronautics and Space Administration Lucy Student Pipeline Accelerator and Competency Enabler Mission Concept Academy PDR Presentation

Doku F., Ekka U., **Venterea R**., Using Quantum Computing to Classify Solar Flares. 2021, HackUTD: The VII Seas

Invited Lectures

Venterea R. C., QuarkNet Workshop: My Academic Journey, 2024, QuarkNet, University of Minnesota, Twin Cities

GRANTS

Past Research

Nicola Tomassetti
Fulbright U.S. Student Program
€ 13,800
October 2024 – July 2025
Developing a Web Application for Cosmic-ray and Space Physics Data
Research Assistant

GBT/24B-184 (John Orlowski-Scherer)
National Radio Astronomy Observatory
August 2024 – December 2024
Investigating Anomalous Flux from the Asteroids (511) Davida and (423) Diotima
Research Assistant

HONORS AND AWARDS

Sons of Italy Foundation National Leadership Grant *2026*

National Science Foundation Graduate Research Fellowship Program *2026 – Present*

Hispanic Scholarship Fund Finalist *2022 – 2024, 2025 – 2026*

Dean's Distinguished Award *2025 – 2026*

JS Enrichment Fund *2024 – 2025*

Foundations of Citizen Science *2024*

Heising-Simons Foundation Grant *2024*

Dean's Distinguished Award, declined *2024 – 2025*

Alpha Phi Alpha Fraternity Memorial Scholarship *2020 – 2024*

Fulbright U.S. Student Program Scholar *2024 – 2025*

Wells Fargo Foundation HSF Scholarship *2023*

Discover Scholar at the University of Chicago *2023*

Grainger Engineering MERGE Scholar *2023*

ASPIRE Illinois Scholar at the University of Illinois Urbana-Champaign *2023*

Dean's List of the College of Arts and Sciences for Excellence in Scholarship *2021, 2023*

Hispanic Scholarship Fund Scholar *2020 – 2021, 2022 – 2024*

Summer Experience Grant *2023*

Undergraduate Research Fund *2023*

Einhorn Discovery Grant *2023*

FOCUS Scholar at the Georgia Institute of Technology *2023*

Society of Hispanic Professional Engineers Undergraduate Scholarship *2022 – 2023*

Inaugural Session of the Nexus Summer Scholars Program *2022*

Meritorious Award for Differential Equations Modeling SIMIODE Challenge *2021*

D. E. Shaw Latitude Fellowship *2021*

National Name Exchange *2021*

COMMUNITY SERVICE

CECI Workshop on Detector Science
Volunteer, June – July 2026
Cleaned, organized posters, and served as presentation timer.

Science x AI Summit
Volunteer, June 2026
Served as receptionist for SAIR participants.

Riverside Unified School District Annual Science & Engineering Fair
Judge, January 2026
Served as a judge for elementary school science fair posters.

UCR Highlander Invitational

Grader, January 2026
Graded Rocks & Minerals Division B of the Science Olympiad tournament.

American Physics Society National Mentoring Community
Mentor, August 2024 – Present
Mentee accepted at MSU, Howard.

Reach the World
November 2024 – April 2025
Described my academic journey abroad to K-12 classrooms.

National Institute of Development Advancement Certified Chapter Leader Program
Facilitator, July 2023

National Academic Quiz Tournaments 2023 High School National Championship Tournament
Full-time scorekeeper, May 2023

Nexus Scholars Program Information Session 2023
Student panelist, October 2022

Zooniverse
Citizen scientist, 2022 – 2023

PROFESSIONAL TRAINING

Fulbright Media Literacy Seminar, Rome, Italy *Mar 2025*
Discussed the implications of artificial intelligence in the global geopolitical sphere.

University of Minnesota, Twin Cities, Minneapolis, Minnesota *Jul 2024 – Aug 2024*
Zwicky Transient Facility Summer School

University of California, Riverside, Riverside, California *Jul 2024 – Aug 2024*
NSF GRFP Application Workshop Series

Coursework
Compute Ontario Advanced Research Computing Training *Jun 2022*
Using JupyterLab

Coursework
Compute Ontario Advanced Research Computing Training *Jun 2022*
Introduction to Python

Cornell University, Ithaca, New York *May 2022 – Jul 2022*
Attended weekly meetings in professional development that emphasized career goals and improving communication and research skills

Coursework
SciNet High Performance Computing Consortium *Jan 2022*
Databases in Scientific Computing

Coursework
SciNet High Performance Computing Consortium *Jan 2022*
Introduction to GPU Programming

Coursework
SciNet High Performance Computing Consortium *Dec 2021*
Intro to SciNet, Niagara, and Mist

Coursework
SciNet High Performance Computing Consortium *Nov 2021*
Intro to the Linux Shell

PROFESSIONAL AFFILIATIONS

Association of Latino Professionals For America *2021 – Present*

Atacama Cosmology Telescope Collaboration *2021 – Present*

Cerro Chajnantor Atacama Telescope Collaboration *2026 – Present*

Cosmic Footprint Society *2024 – Present*

National Italian American Foundation *2025 – Present*

National Society of Hispanic Physicists *2021 – Present*

Order Sons and Daughters of Italy in America *2025 – Present*

Society for the Advancement of Chicanos and Native Americans in Science *2023 – Present*

Society of Hispanic Professional Engineers *2021 – Present*

COMPUTER SKILLS

Programming: Python, MATLAB, R, Java, LaTeX, Unix shell, JavaScript, HTML, CSS

Applications: GitHub, Jupyter Notebook, JupyterLab, Google Colab, ArcGIS, JMARS, Visual Studio Code, Google Docs Editors, Microsoft Office, MobaXterm, SigmaPlot, Zotero, Amazon AWS

EXTRACURRICULARS

Future GRADS MentorSHPE *2023 – 2024*

Mentoring program for prospective graduate students from underrepresented backgrounds emphasizing networking and writing graduate application essays. Hosted by the Society of Hispanic Professional Engineers.

SIMIODE Challenge Using Differential Equations Modeling VI *2021*

Modeled the spread and censorship of misinformation on social media platforms via differential equations with an epidemiological approach. Presentation won Meritorious Award.

Directed Reading Program *2021*

Learned topics in quantum field theory. Learned topics in differential geometry. Presented on both subject areas.

HackUTD: The VII Seas *2021*

Collaborated in a team of three to implement a quantum computer algorithm to classify solar flare data. Successfully implemented with an accuracy of 98% 24 hours in advance.

Cornell Society of Physics Students *2020 – 2024*

Cornell Math Club *2020 – 2021*

Cornell Astronomical Society *2020 – 2024*

Cornell Chess Club *2020 – 2024*

Competed in the 2023 – 2024 United States Amateur East Championships. Captain of the Cornell C Team.

Cornell Mathematical Contest in Modeling *2020*

Helped develop an algorithm to optimize scheduling for the Food Bank of the Southern Tier.

Cornell Undergraduate Research Board *2020*

Mentoring program for undergraduate students planning research careers emphasizing professional development, presenting research, and interacting with professors.

LANGUAGES

English: Native Language

Italian: Intermediate in Listening, Speaking, Reading, and Writing

HOBBIES

Learning, stargazing, playing classical music, reading, fishing for small mouth bass, badminton, brewing coffee